MPG No. 12.3.42 4′22′ (4’22’ setting) [Type III, tetragonal]

Table 1: Wyckoff site: `1o`, site symmetry: `4’22’`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8]`

Table 2: Wyckoff site: `2a`, site symmetry: `4’..`

No.	position	mapping
1	$[0, 0, z]$	`[1,4,6,7]`
2	$[0, 0, -z]$	`[2,3,5,8]`

Table 3: Wyckoff site: `4b`, site symmetry: `..2’`

No.	position	mapping
1	$[x, x, 0]$	`[1,8]`
2	$[-x, -x, 0]$	`[4,5]`
3	$[-x, x, 0]$	`[3,6]`
4	$[x, -x, 0]$	`[2,7]`

Table 4: Wyckoff site: `4c`, site symmetry: `.2.`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$[-x, 0, 0]$	`[3,4]`
3	$[0, x, 0]$	`[6,8]`
4	$[0, -x, 0]$	`[5,7]`

Table 5: Wyckoff site: `8d`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[4]`
3	$[-y, x, z]$	`[6]`
4	$[y, -x, z]$	`[7]`
5	$[-x, y, -z]$	`[3]`
6	$[x, -y, -z]$	`[2]`
7	$[y, x, -z]$	`[8]`
8	$[-y, -x, -z]$	`[5]`